

Dual Governance for Continual Agents: Separating Reversible Information Acquisition from Persistent Self-Modification

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Abstract

A continual agent repeatedly chooses whether to gather information and whether to change persistent structure. Applying one evidence threshold to both actions is a category error: sensing is often reversible and utility-driven, whereas a structural mutation can propagate indefinitely and requires lifetime-valid authority. We formalize a dual-governance interface that assigns typed capabilities to evidence. A posterior utility controller decides whether a binary cue is worth its cost; its record cannot authorize a model mutation. Persistent release instead requires disjoint discovery, structural, semantic, and resource streams, a preallocated lifetime grant, a predictable confirmation process, and an exact hash-linked witness. Three registered reference studies isolate this interface. The utility controller attains 0.00556 mean regret on a stationary grid, versus 0.03376 for an exact per-task test. Reusing reversible support as structural evidence escalates in 2,000/2,000 null lifetimes, while the typed firewall crosses in 63/2,000 (3.15%; Wilson upper 4.01%) under its declared 5% family budget and rejects 112/112 substitutions. A split-budget learned refinement interval controls its registered lifetime boundary while an unbudgeted plug-in point falsely escalates both permitted refinement boundaries in 1,000/1,000 lifetimes. These results validate a narrow governance lifecycle for supplied binary mechanisms, not autonomous structural discovery. **Keywords:** continual learning, active sensing, e-processes, lifetime error, structural adaptation, capability separation

1 Introduction

An adaptive agent makes actions with different reversibility. Buying one observation, retaining a temporary context, updating a belief, splitting a persistent state, and adding a mechanism should not inherit the same authority from the same statistic. A conservative threshold suitable for an irreversible mutation can waste useful information; a utility rule suitable for sensing can create repeated false structural changes if relabelled as scientific evidence.

Sequential testing supplies anytime-valid evidence [2–4], while Bayesian decision rules naturally price information. The contribution here is not a new theorem in either area. It is a typed system interface that prevents evidence

earned for one action from escalating into authority for another.

We study three questions. First, how costly is a structural-style threshold when the true decision is a reversible information purchase? Second, can a combined lifecycle prevent positive support evidence from authorizing a persistent mutation under fresh null confirmation? Third, can a refinement null learned on discovery data be frozen and charged without silently using the structural error budget twice?

2 Action types and authority

Let an action have descriptor

$$\tau(a) = (\text{reversibility, persistence, scope, rollback, externality}).$$

The authority function $\mathcal{A}(e, \tau)$ maps an evidence record e to the actions it permits. A valid record is therefore not a bearer token. It binds the action type, data occurrences, candidate, query, null, protocol, and ledger state.

For a reversible cue, support agreement satisfies $A_i \mid \rho \sim \text{Bernoulli}(\rho)$. A cue costs c , succeeds with probability ρ , and is compared with no-cue success $1/2$. Its expected value is positive exactly when

$$\rho > \frac{1}{2} + c.$$

A Beta posterior utility controller probes when

$$\mathbb{E}[\rho \mid A_{1:n}] > \frac{1}{2} + c.$$

This is a decision rule, not a structural-validity certificate.

By contrast, investigation i for a persistent action receives a predictable grant such as

$$\alpha_i = \frac{\alpha}{i(i+1)}, \quad \sum_i \alpha_i \leq \alpha.$$

The grant is irrevocably spent and bound to the investigation context. A predictable confirmation e-process may then control a lifetime union via Ville’s inequality and a union bound, provided its filtration and null assumptions hold.

Table 1: Mean expected regret on the stationary reliability-cost grid.

Controller	regret
Posterior utility	0.005558
Exact per-task test	0.033759
Mixture e-value	0.048811
Always probe	0.027917
Never probe	0.118750
Oracle	approximately 0

3 Typed causal lifecycle

The combined lifecycle has four disjoint stages:

1. discovery proposes a candidate and may include reversible support;
2. structural confirmation tests fresh behaviour;
3. semantic confirmation tests declared consequence queries; and
4. resource confirmation tests the deployment cost and reliability scope.

The reversible record binds support occurrence IDs, agreement stream, fixed windows, and its statistic. Its only capability is `probe`; it always has `authoritative_library_changed=false`. Persistent confirmation requires pairwise-disjoint occurrence sets and a grant launched before context freezing.

Before confirmation, a witness freezes the exact candidate snapshot, producer, confirmation IDs, and reliability grid. The candidate predicts before each outcome. At release, the agreement stream and e-path are recomputed from that witness. Candidate, producer, query, stream, or path substitution fails.

Typed persistent actions include recalibration (no mechanism-library change), state split (semantic evidence required), and mechanism admission (fresh semantic and structural evidence). Passing a reversible record to a persistent API is a type error.

4 Study 1: reversible sensing

The stationary benchmark crosses six reliabilities with four costs. Controllers are never probe, always probe, posterior-mean utility, an exact binomial test, a mixture likelihood-ratio e-value, and an oracle. Temporal conditions include abrupt decay, abrupt recovery, gradual decay, and a query-only shift invisible in support data. Global, trailing-window, and Bayesian online change-point estimators are compared [1].

The posterior controller has 16.46% of the exact test’s regret. Across 2,000 streams of 256 least-favourable null tasks for each of four seeds, family incidence of at least one probe is 0.99975 for an uncorrected exact test and 0.031375 for summable grants. The posterior controller probes in every lifetime; that is expected because its actions are utility decisions, not family-wise claims.

After abrupt decay, Bayesian changepoint inference reduces regret from 0.099425 to 0.018425 relative to the global posterior; after abrupt recovery, from 0.270200 to

0.006738. A trailing eight-observation window is still better at 0.013525 and 0.002275. Under a query-only shift from support reliability 0.95 to query reliability 0.5, all support-only adaptive controllers probe at least 99.125% of the time and incur about 0.1 regret. No estimator repairs absent information.

5 Study 2: escalation firewall

For fresh binary confirmation agreements a_t , the structural process is a symmetric directional mixture likelihood ratio:

$$E_t = \frac{1}{2|G|} \sum_{p \in G \cup (1-G)} \prod_{s=1}^t \frac{p^{a_s}(1-p)^{1-a_s}}{(1/2)^{a_s}(1/2)^{1-a_s}}.$$

Under fresh IID fair-branch confirmation, it is a nonnegative martingale. The benchmark uses 2,000 lifetimes of 128 investigations. A negative control has strong reusable support but fair structural confirmation. A naive system reuses the support statistic; the firewall recomputes only from fresh confirmation.

The firewall has 63/2,000 null-crossing lifetimes (3.15%, Wilson 95% upper 4.0098%), whereas naive support escalation crosses in all 2,000. Fresh alternative power is 99.9957%, and reversible probe power is 99.9316%. Eight integrated alternative lifecycles release; eight null lifecycles deny; all 112 registered candidate, query, stream, path, witness, and escalation substitutions are rejected; and no reversible decision mutates the library.

The immutable artifact hash is 06D3F7D51C46D632...AC0417; the independent reconstruction passes 15 checks and has science hash F3C8BCB7CE00553B...EED80A9.

6 Study 3: learned refinement null

The preceding study receives its reusable null from the experimenter. We next estimate a Bernoulli persistence interval only from fixed-size discovery data. For investigation i , the total structural budget 0.05 is split into two telescoping ledgers:

$$\delta_i = \frac{0.025}{i(i+1)}, \quad \gamma_i = \frac{0.025}{i(i+1)}.$$

With K_i successes among n discovery observations,

$$\hat{p}_i = K_i/n, \quad r_i = \sqrt{\frac{\log(2/\delta_i)}{2n}},$$

and $C_i = [\max(0, \hat{p}_i - r_i), \min(1, \hat{p}_i + r_i)]$. For registered refinement radius $\tau = 0.05$, confirmation tests the envelope

$$[L_i, U_i] = [\max(0, \inf C_i - \tau), \min(1, \sup C_i + \tau)].$$

If the envelope touches 0 or 1, the implementation abstains.

Hoeffding gives $P(p_i \notin C_i) \leq \delta_i$. Conditional on coverage, the fresh predictable interval process crosses falsely

with probability at most γ_i . A union bound therefore controls the declared lifetime by $\sum_i(\delta_i + \gamma_i) \leq 0.05$. This does not cover dependent or optionally stopped discovery.

Across 1,000 lifetimes of 64 investigations, discovery coverage fails in 1/1,000 lifetimes. Learned centre, lower, upper, and adaptive-null incidences are 0.10%, 0.10%, 0.20%, and 0%; the largest Wilson upper bound is 0.7263%. An unbudgeted plug-in point escalates both permitted refinement boundaries in 1,000/1,000 lifetimes. Learned-interval high/low task power is 99.9953%/100%. Mean and maximum interval width are 0.13628 and 0.13940. All integrated alternatives release, boundary nulls deny, and all 80 substitutions are rejected.

The artifact hash is F5C9307A3E8C0E59...1C8767F; its independent audit passes 20 checks with science hash 4AC4D1947C78F399...8B3739F.

7 Discussion

The three studies support one architectural distinction. Information actions should be chosen by expected value under their reversibility and cost. Claims that authorize persistent change need fresh, scoped, lifetime-accounted evidence. Evidence records require typed capabilities so a positive result cannot be replayed across a semantic boundary.

The result also identifies a hierarchy of missing work. Binary agreement is not a learned nonlinear world model. Fixed IID discovery is not a dependent online stream. A learned interval around one Bernoulli persistence parameter is not a functional confidence set. Integrity hashes detect registered substitution but do not prove arbitrary program semantics. Finally, lifetime error control does not show that the ontology or consequence-query class is adequate.

8 Limitations

The mechanisms, queries, and filtration are supplied. The first study uses synthetic Bernoulli support. The firewall null is IID fair branching. The learned-null study uses fixed-size IID discovery and a scalar refinement envelope. There is no learned representation, natural-domain novelty proposal, dependent-noise theorem, hardware experiment, or external reproduction. High power against separated alternatives should not be extrapolated to small effect sizes.

9 Conclusion

Reversible sensing and persistent self-modification are different control problems. A continual agent can use permissive utility decisions for routine information gathering while reserving lifetime-valid, fresh, and scoped evidence for persistent structure. The next decisive step is to learn nonlinear refinement and cause-set proposals on disjoint data, test them under dependent streams, and embed the firewall in a learned causal world model.

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